# Lucas A. **Saavedra**

Computer Engineer · AI Researcher

*Buenos Aires, Argentina*

lucasarielsaavedra@hotmail.com | lucassaavedra123.github.io | lucassaavedra123 | lucassaave

## Experience

### Accenture

*Buenos Aires, Argentina*

Senior Full-Stack AI Developer — *May 2025 - PRESENT*

- GenAI Studio team: LLM-based multi-agent systems (LangGraph, FastAPI, Azure OpenAI, Elasticsearch) for automated code migration between languages, data-quality testing, and AI-assisted accounting reconciliation.
- Social media analytics platform: Neo4j knowledge graph, Spring Boot batch classification pipeline, and a LangGraph conversational agent exposed through a Next.js chat interface, including an MCP server to expose the graph as tools for the agent.
- Real-time computer vision systems (YOLO, OpenCV, TensorFlow) for vehicle detection, customer profiling, and dataset preparation for gas station clients.
- Full-stack development with Next.js/React and enterprise Angular applications, deployed with Docker and Azure Pipelines.
- Research on Small Language Models (SLMs) and development of agents to automate the Software Development Life Cycle (SDLC) through Spec-Driven Development.
- **Technologies used: Python, Large Language Models (LLMs), LangGraph, FastAPI, Azure OpenAI, ElasticSearch, Neo4j, Spring Boot, Next.js, React, Angular, Docker, Azure Pipelines, Computer Vision (YOLO, OpenCV, TensorFlow).**

### BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

*Buenos Aires, Argentina*

Research Assistant — *Jan. 2022 - PRESENT*

- Implemented Deep Learning models for static and dynamic analysis of the nicotinic acetylcholine receptor (nAChR), with orientation to graph theory, recurrent networks and geometric computing.
- Analyzed distribution and density of nicotinic acetylcholine receptors in the plasma membrane using super-resolution microscopy (STED, STORM), with advanced analytics implemented in MATLAB based on graph theory and computational geometry.
- Developed an efficient and autonomous algorithm based on supervised Graph Neural Networks (GNNs) in TensorFlow to detect and quantify dynamic particle clustering in single-molecule microscopy (SMLM) data.
- Applied AlphaFold3-equivalent models to study protein-ligand structures (nAChR and cholesterol), combined with molecular dynamics (MD) simulations using GROMACS on high-performance computing (HPC) platforms with SLURM.
- Developed WadTCN, a deep learning architecture for 2D single-particle diffusion analysis (anomalous exponent and diffusion coefficient estimation, model classification, trajectory segmentation), applied to nAChR trajectories in live cells.
- Analyzed two-colour sub-millisecond MINFLUX single-molecule tracking data of nAChR and cholesterol in live cells, with trajectory segmentation and diffusive-regime classification.
- Design and programming in C++ of electronic circuits with Arduino to control piezo motors.
- **Technologies used: Python, Java, C++, TensorFlow, PyTorch, Pandas, NumPy, Docker, MongoDB, MATLAB, GROMACS, SLURM, Arduino.**

### Data Science and AI Laboratory, Pontificia Universidad Católica Argentina (UCA)

*Buenos Aires, Argentina*

Research Assistant — *Jan. 2025 - Jan. 2026*

- Supervisor: Prof. Antonio Quintero-Rincón.
- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning—specifically, hierarchical variational autoencoders—to detect specific cognitive activity with PyTorch and MNE-Python.

### Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

*Buenos Aires, Argentina*

Assistant Professor — *Mar. 2023 - June 2025*

- ”Introduction to Data Science”: Artificial Intelligence concepts, Neural Networks, Scientific Method.
- ”Data-Oriented Programming”: Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- ”Discrete Mathematics”: Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

### UpWork

Freelancer — *Dec. 2023 - May 2025*

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2 Linux instances.
- YOLO training to identify basketball players in video matches from a European basketball league, with a SQL database hosted in Supabase.
- GUIs developed with built-in Napari plugins to gather information from laser spot images, combined with decision tree classifiers in scikit-learn.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium, Napari, scikit-learn.**

### Deloitte

*Buenos Aires, Argentina*

Back-End Developer — *Jun. 2021 - Jun. 2022*

- Developed Codex Gigas, a malware ”DNA” search engine that identifies patterns and characteristics for malware hunting usign REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

**Deloitte** *Buenos Aires, Argentina*

Cyber Risk Advisory Intern *Jan. 2021 - Jun. 2021*

- Offensive/Defensive Security and Threat Intelligence basic concepts.

## Skills

| | |
|---|---|
| **DevOps** | AWS, Docker, CircleCI |
| **Technologies** | Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis, Selenium, Docker, YOLO, Langgraph, Langchain, Azure OpenAI |
| **Programming** | Python, JAVA, C, C++, R, LaTeX, MATLAB |

## Education

**Pontificia Universidad Católica Argentina (UCA)** *Buenos Aires, Argentina*

Computer Engineering *2018 - 2023*

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)
- **GPA:** 8.02/10.

## Publications and conferences

| | | |
|---|---|---|
| 2022 | **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences | |
| 2023 | **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium | *Punta del Este, Uruguay* |
| 2024 | **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale | |
| 2024 | **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0 | |
| 2025 | **Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis**, Journal of Physics: Photonics | |
| 2025 | **Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking**, Nature Communications | |
| 2025 | **Quantitative evaluation of methods to analyze motion changes in single-particle experiments**, Nature Communications | |
| 2025 | **Environmental enrichment delays the onset of cognitive decline in a transgenic mouse model of Alzheimer disease**, XL Annual Meeting SAN | |
| 2026 | **Machine Learning in Single-Molecule Tracking Analysis of Superresolution Optical Microscopy Data**, Cells | |
| 2026 | **A self-supervised reconstruction framework for segment-level detection of task-induced deviations in multi-channel EEG**, PLOS ONE (Under review) | |

## Certifications and Courses

2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
2020 **CS50's Introduction to Artificial Intelligence with Python**, Harvard, edX
2021 **Machine Learning**, Stanford Online, Coursera
2021 **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
2021 **R Programming**, Johns Hopkins University, Coursera
2021 **Docker for the Absolute Beginner - Hands On - DevOps**, Udemy
2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
2022 **AWS Certified Cloud Practitioner**,
2022 **Data Science**, Coderhouse
2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
2025 **Introduction to Statistics**, Stanford Online, Coursera
2025 **Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)**,
2025 **Bioinformatics; Learn Docking & Mol Dynamics Simulation**, Udemy
2025 **Basic to Advanced: Retrieval-Augmented Generation (RAG)**, Udemy
2025 **IELTS Score: 7.5**,
2026 **Golang for Beginners**, Udemy
2026 **Kubernetes for the Absolute Beginners - Hands-on**, Udemy